

Differentiable, Physics-Informed 3D Gaussian Splatting:

Unified Paradigms for Kinematic Evolution, Inverse Simulation, and Functional Design

Source guide

This text describes a technological evolution in computer vision that merges 3D Gaussian Splatting with computational mechanics to create scenes that are both visually realistic and physically accurate. By treating rendering primitives as Lagrangian material points, researchers have developed a bidirectional framework where visual data informs physical properties like mass and stress, while physical laws stabilize the reconstruction of dynamic objects. This synergy allows for inverse simulation, enabling the estimation of hidden material traits from video, and functional design, where objects are optimized to meet specific aerodynamic or fluid goals. Ultimately, the source highlights how physics-informed paradigms overcome traditional limitations like visual artifacts and occlusions, paving the way for more robust digital twins and robotic interactions.

The virtualization of real-world environments has experienced a paradigm shift with the transition from continuous implicit scene representations, such as Neural Radiance Fields, to explicit, unstructured optimization frameworks.[1, 2] 3D Gaussian Splatting (3DGS) has emerged as a dominant methodology for real-time, photorealistic novel view synthesis, utilizing millions of learnable anisotropic Gaussian primitives optimized via differentiable rasterization.[1, 3]

However, classic 3DGS is fundamentally limited by its lack of physical awareness.[4, 5] Traditional formulations treat scene reconstruction as a purely visual image-fitting problem, optimizing geometric shapes without considering the physical principles governing dynamics, material behavior, or conservation laws.[1, 4, 5] Consequently, under sparse views, temporal occlusions, or novel external forces, purely data-driven dynamic 3DGS fails to maintain physical consistency, resulting in floating artifacts, spatial divergence, and geometric collapse.[1, 6, 7] To overcome these visual limitations, modern computer vision and computational mechanics have converged on differentiable, physics-informed 3DGS.[8, 9, 10] This framework establishes a bidirectional relationship: rendering primitives are endowed with physical properties (such as mass, velocity, strain, and stress) to drive forward physical simulations, while differentiable rendering losses backpropagate gradients through physical solvers to estimate material properties, identify forces, and optimize functional shapes.[4, 8, 10]

Mathematical Foundations of Splat-Particle Duality

The core innovation of physics-integrated 3DGS is the mathematical mapping between visual rendering primitives and Lagrangian material points, eliminating the need for auxiliary geometric wrappers such as tetrahedral meshes, cage meshes, or marching cubes. [11, 12, 13] A dynamic scene is discretized into a set of spatially localized 3D Gaussian kernels $g_{\{i\}}$, each defined by a center position $\mu_{\{i\}} \in \mathbb{R}^3$, an anisotropic covariance matrix $\Sigma_{\{i\}} \in \mathbb{R}^{3 \times 3}$, an opacity $o_{\{i\}} \in [0, 1]$, and spherical harmonics coefficients $h_{\{i\}}$ for view-dependent color. [1, 8] The covariance matrix is parameterized by a scaling diagonal matrix $S_{\{i\}} = \text{diag}(s_{\{i\}})$ and a rotation matrix $R_{\{i\}}$ governed by a unit quaternion $q_{\{i\}}$:

$$\Sigma_{\{i\}} = R_{\{i\}} S_{\{i\}} S_{\{i\}}^{\top} R_{\{i\}}^{\top} \quad [3]$$

In physics-integrated pipelines, such as PhysGaussian and its derivatives, these visual ellipsoids serve directly as the discretized particles of a continuum mechanics simulation. [11, 14] Under this formulation, each Gaussian particle is assigned physical properties including mass $m_{\{i\}}$, reference density $\rho_{\{i\}}$, volume $V_{\{i\}}$, velocity $v_{\{i\}} \in \mathbb{R}^3$, and a deformation gradient $F_{\{i\}} \in \mathbb{R}^{3 \times 3}$. [8, 11, 15] To maintain physical consistency during dynamic deformations, the spatial covariance matrix of each Gaussian is updated using the first-order kinematics of the deformation gradient [11, 13]:

$$\Sigma_{\{i,t\}} = F_{\{i,t\}} \Sigma_{\{i,0\}} F_{\{i,t\}}^{\top} \quad [16]$$

where $\Sigma_{\{i,0\}}$ is the initial, undeformed rest-state covariance. [16] The spatial orientation and stretching of the visual primitive are thus coupled with the underlying mechanical strain. [11, 14] is the initial, undeformed rest-state covariance.[16] The spatial orientation and stretching of the visual primitive are thus coupled with the underlying mechanical strain.[11, 14]

Continuum Mechanics and Material Point Discretization

To model complex material behaviors—ranging from elastic jellies and plasticine to granular sands and viscoplastic fluids—the mechanical state updates are governed by the conservation of mass and momentum. [11, 14] Mass conservation is naturally satisfied by tracking constant Lagrangian mass $m_{\{i\}}$ for each particle. [11] Momentum conservation is computed by solving the Cauchy momentum equation:

$$\rho (\partial v / \partial t + v \cdot \nabla v) = \nabla \cdot \sigma + \rho g \quad [1, 11]$$

where σ is the Cauchy stress tensor and g represents external body forces such as gravity. [1, 11, 17]

The Material Point Method (MPM) is widely used to solve this partial differential equation. [8, 11, 18] MPM interpolates physical state variables between Lagrangian particles and a background Eulerian grid using C^1 -continuous B-spline kernels $w_{\{ip\}}$. [11, 18] This particle-to-grid transfer computes grid mass $m_{\{i\}}^n$ and grid velocity $v_{\{i\}}^n$ at time step t^n :

$$m_{\{i\}}^n = \sum_{\{p\}} m_{\{p\}} w_{\{ip\}}^n, \quad v_{\{i\}}^n = (1 / m_{\{i\}}^n) \sum_{\{p\}} m_{\{p\}} v_{\{p\}}^n w_{\{ip\}}^n \quad [11]$$

The divergence of stress is computed on the background grid to determine nodal forces and accelerations, which are then used to update the velocities and positions of the Lagrangian Gaussian particles.[11, 18]

Feature / Formulation	Explicit MPM (e.g., PhysGaussian) [11, 18]	Implicit MPM (e.g., i-PhysGaussian) [18, 19]	Learnable Momentum Fields (e.g., PIDG) [1]
Time-Integration Scheme	Forward Euler [11]	Backward Euler / Implicit Newton [18, 20]	Time-evolving Neural Material Field [1]

Stability Limitation	Bounded by Courant-Friedrichs-Lewy (CFL) condition [18, 19]	Unconditionally stable (up to 20× larger time steps) [18, 19]	Regularized by neural network priors and Cauchy residuals [1]
Solver Type	Step-by-step algebraic updates [18, 19]	Matrix-free preconditioned GMRES [18, 19]	Optimization of PDE residual loss terms [1]
Primary Limitation	Instability with high-stiffness materials or large Δt [18, 19, 20]	High computational cost per solver step [18, 19]	Requires dense physical training datasets [1]
Target Materials	Viscoplastic fluids, granular soils, low-stiffness solids [11, 14]	High-stiffness elastic solids, quasi-static deformations [18, 19, 20]	Complex fluids and generalized elastic systems [1]

The Implicit Newton-GMRES Formulation

While explicit MPM schemes are efficient per time step, they suffer from step-size sensitivity. [18, 19] Under high material stiffness (such as metals or rigid rubber) or quasi-static movements, explicit integrators degrade in accuracy or diverge unless an impractically small time step Δt is used. [18, 19]

To address this, i-PhysGaussian replaces the explicit formulation with an implicit MPM integrator. [18, 19] The end-of-step velocity v_{i}^{n+1} on the grid is computed by minimizing a momentum-balance residual vector $R(v_{i}^{n+1})$:

$$R(v_{i}^{n+1}) = m_{i} (v_{i}^{n+1} - v_{i}^{n}) - \Delta t (f_{i}^{\text{int}}(x_{i}^{n+1}) + f_{i}^{\text{ext}}) = 0 \quad [18, 20]$$

where f_{i}^{int} and f_{i}^{ext} represent the internal and external forces acting on grid node i , respectively. [18, 20] This non-linear system is solved using an implicit Newton-type optimization outer loop. [18, 19] Due to the high dimensionality of the grid space, forming the Jacobian matrix directly is computationally prohibitive. [18, 19] i-PhysGaussian therefore employs a right-preconditioned, matrix-free Generalized Minimal Residual (GMRES) inner loop to compute Newton descent directions, which keeps the simulation stable at up to 20× larger time steps than explicit alternatives. [18, 19]

Differentiable Coupling and Gradient Propagation Pathways

The integration of physical solvers with 3DGS allows gradients to flow end-to-end through both the rendering and simulation pipelines.[8, 9, 21] This enables inverse graphics, where 2D

image-space losses are backpropagated to estimate the underlying physical parameters of the scene.[9, 10, 22]

Control-Space Guidance and Trajectory Stabilization

Directly backpropagating visual gradients to update Lagrangian particle positions x often introduces high-frequency visual noise into the physical simulation, causing numerical instability and geometric deformation. [16] This instability occurs because the rendering gradients $\partial L_{\text{render}}/\partial x$ represent localized, pixel-level adjustments that can conflict with the conservative, low-frequency forces of continuum mechanics. [16]

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To stabilize this optimization, PhysMorph-GS routes the rendering gradients through the deformation gradient F rather than particle positions. [16] The spatial gradient components are zeroed ($\partial L_{\text{render}}/\partial x = 0$), and the visual feedback is injected exclusively as a penalty on the deformation gradient $\partial L_{\text{render}}/\partial F$. [16]

This formulation converts 2D multi-view silhouette and depth supervision into a smoothed control-space force.[16] The optimization pipeline runs in three distinct stages:

1. Physics Rollout: The MPM forward simulation runs for T timesteps. [16] A learnable control deformation field δF_{c} is optimized using the Adam optimizer to minimize an end-layer mass loss on the Eulerian grid relative to the target mass distribution. [16] The render gradient $\partial L_{\text{render}}/\partial F$ from the previous frame is injected as a penalty on the deformation gradient F . [16]
2. Multi-View Rendering and Gradient Extraction: The final particle states (x_{T} , F_{T}) are rendered from up to $V=8$ ring-placed cameras using the differentiable $F \rightarrow \Sigma$ mapping. [16] Rendering losses—including distance transform, soft intersection-over-union (IoU), and depth—are computed against target silhouettes. [16] Backpropagating through the rendering pipeline yields $\partial L_{\text{render}}/\partial F$, which is stored for the next frame. [16]
3. Chamfer Plasticity Injection: The render gradient provides lightweight guidance through the deformation gradient from the first frame. [16] To refine fine geometric details, Chamfer-guided plasticity is activated after a warmup period of k_{0} frames once coarse structural seeds have formed. [16]

Monocular Visual-Physical Bridges

Estimating 3D physical parameters from a single camera view is highly underconstrained due to scale ambiguity, inaccurate geometry, and weak coupling between appearance optimization and physical simulation.[15, 23, 24] Standard 3DGS rasterization only carries gradients through color and opacity, which provides weak positional feedback for geometry optimization.[15]

To address these limitations, MonoPhysics introduces three visual-physical bridges to enable monocular inverse physics optimization [15, 25]:

- Global Scale Alignment: Scale ambiguity is resolved by introducing a single learnable scalar parameter that reparameterizes the global coordinate frame, aligning the monocular camera's relative coordinates with physical units.[15, 25]
- Physics-Aware Geometric Refinement: Instead of keeping the initial geometry frozen, MonoPhysics computes per-particle volumes from the spatial distribution of the Gaussians.[15] It then relocates the primitives using a combination of visual rendering

losses and physical simulation feedback, allowing the geometry to adapt dynamically.[15]

- Differentiable Position Map: To provide direct spatial gradients, MonoPhysics implements a differentiable position map.[15, 25] This map defines a direct gradient path from image-space silhouette alignment and pixel-location losses back to the 3D particle positions $\mathbf{x}$, bypassing the visual limitations of standard 3DGS rasterization.[15]

To handle complex non-linear deformations, MonoPhysics fits a small implicit neural network that encodes spatially varying weights.[23] These weights act as a reduced deformation basis, which is trained by evaluating elastic energy changes via stochastic Monte Carlo sampling of the deformation volume.[23]

Multi-Body Interactions, Occlusions, and Articulated Systems

A major challenge in dynamic scene reconstruction is maintaining object tracking when objects are temporarily occluded.[6, 22] In purely visual dynamic 3DGS, once an object is occluded from all training views, photometric gradients vanish.[6, 22] Without gradient supervision, the Gaussians representing the object prune, drift, or collapse, preventing the model from maintaining object permanence.[22, 26]

Differentiable Rigid-Body Simulation for Occlusion Recovery

PersistGS addresses this limitation by coupling differentiable rigid-body simulation with 3DGS to maintain object permanence during occlusions.[6, 22, 27] The framework decomposes the scene into per-object Gaussians and explicit collision meshes.[6, 22, 27]

During visible frames, the object's pre-occlusion trajectory is used to estimate physical parameters, including initial velocity and surface friction, via an observability-aware curriculum.[6, 22, 27] During occlusions, a differentiable rigid-body simulator predicts the object's $SE(3)$ trajectory.[6, 22, 27] Unlike kinematic linear extrapolation, this physics-based trajectory naturally models contact interactions, boundary bounces, and friction deceleration.[22, 27]

To optimize the physical parameters, PersistGS introduces a centroid silhouette loss that decouples positional tracking from high-frequency appearance noise.[22, 27] This loss computes the centroid of the rendered object mask relative to the target silhouette, reducing trajectory estimation error by 40% compared to standard photometric losses.[22, 27]

Articulated Kinematics and Multi-Body Systems

Beyond unconstrained rigid motion, many real-world interactions involve articulated objects with localized physical constraints, such as hinges or joints.[28] ArtGS integrates visual-physical modeling into 3DGS for articulated object understanding.[28]

The pipeline begins by reconstructing static 3D Gaussians from multi-view RGB-D inputs. [28] A Vision-Language Model (VLM) is then used to parse the scene, identify articulated parts, and initialize the structural bones of the skeleton. [28] During dynamic interactions, ArtGS uses differentiable rendering to optimize the joint parameters—including hinge axis directions and origins—under explicit physical constraints. [28] This visual-physical coupling preserves spatial and temporal continuity, allowing for accurate joint estimation even under small deformations (such as hinge rotations of less than 10° or displacements of less than 10 cm) where purely visual point-cloud baselines degrade. [28]

For elastic multi-body interactions, Spring-Gaus integrates a 3D Spring-Mass model directly into the 3D Gaussian primitives.[9, 29, 30] The system extracts a set of structural anchor points from

the reconstructed Gaussians and binds them with virtual spring networks.[30, 31] During optimization, the spring forces govern particle motion, while an online refinement step updates the Gaussian parameters to prevent appearance drift and maintain rendering quality.[30, 31]

Visual Representation	Physical Model	Motion Constraints	Differentiability Pathway	Target Phenomenon
PersistGS [22, 27]	Rigid Body Simulation [6, 22]	SE(3) Trajectory, Contact & Friction [22, 27]	Centroid Silhouette Loss [22, 27]	Rigid motion under complete visual occlusion [22]
ArtGS [28]	Articulated Skeleton [28]	Hinge joints, Axis Direction & Origin [28]	Differentiable Rasterization of skeleton bones [28]	Robotic manipulation of articulated objects [28]
Spring-Gaus [9, 29]	3D Spring-Mass [9, 30]	Elastic spring networks, anchor point binding [30, 31]	Direct backpropagation from image frames to spring constants [9, 29]	Multi-view elastic deformation tracking [9, 29]

Neural Fields, Learned Operators, and Functional Optimization

While grid-based MPM and explicit spring-mass simulators offer high physical fidelity, they can be computationally expensive, which limits their use in real-time, closed-loop applications.[32] To improve performance, several frameworks replace traditional physics engines with learned neural operators and field representations.[1, 32]

Learned Force Fields and Momentum Models

The Neural Gaussian Force Field (NGFF) framework implements an end-to-end neural simulator for dynamic scenes.[32] NGFF uses a feed-forward geometry transformer to reconstruct 3D Gaussians from multi-view RGB images.[32] The Gaussians are then encoded into high-dimensional feature vectors and processed by a DeepONet-based neural operator.[32] This operator predicts object-centric force fields, which are integrated using an Ordinary Differential Equation (ODE) solver to simulate complex multi-object interactions.[32] This learned formulation avoids grid-based computations, achieving a 100× speedup over traditional simulators.[32]

In monocular view synthesis, Physics-Informed Deformable Gaussian Splatting (PIDG) treats each Gaussian as a Lagrangian material point governed by a time-evolving material field.[1] This material field predicts velocity and constitutive stress for each particle.[1, 33] The framework enforces physical laws by penalizing the residuals of the Cauchy momentum equation.[1]

To supervise the simulation, PIDG introduces a Lagrangian particle flow matching loss.[1] This loss projects the 3D Gaussian velocities onto the 2D image plane to match camera-compensated optical flow, constraining the learned material field to match observed physical motion:

L_{flow}

[1]

Functional Design and Aerodynamic Optimization

In addition to reconstructing observed dynamics, the differentiability of the visual-physical pipeline enables functional shape optimization.[10] Physics-Guided 3D Gaussian Splatting (PG-3DGS) optimizes 3D structures to satisfy specific physical functionalities from 2D observations.[10]

To bridge the visual representation and the physical simulator, PG-3DGS constructs a soft occupancy solid mask χ on a voxelized simulation grid.[10] This mask is defined as a differentiable function of the Gaussian parameters:

$\chi(\mathbf{x}; \theta)$ [10]

where θ represents the positions, scales, rotations, and opacities of the Gaussians.[10] This differentiable coupling allows gradients from complex PDE solvers (such as Navier-Stokes fluid simulations or aerodynamic lift solvers) to flow directly back to the Gaussian parameters.[10]

During optimization, the total loss function balances visual appearance matching L_{vis}

with a specified physical objective function L_{phys}

$= \Phi(\tau)$, where τ represents the simulated trajectory of the physical system [10]:

$J(\theta) = L_{\text{vis}}$

$+ \lambda L_{\text{phys}}$

[10]

This joint optimization has been demonstrated on several functional design tasks [10]:

- Fluid Flow Engineering (Teapots): PG-3DGS optimizes teapot geometries to ensure a connected internal cavity and spout, enabling successful fluid pouring while maintaining visual realism.[10, 34]

- **Aerodynamic Airfoil Design:** The framework modifies wing profiles (e.g., for a Cessna, B-2 Spirit, or paper plane) to maximize aerodynamic lift under uniform airflow, producing geometries that generate higher scale-measured lift in wind-tunnel tests compared to purely visual baselines.[10, 34]

Bidirectional Physics-Vision and Inverse Problems

The coupling of visual and physical representations enables a bidirectional relationship.[4, 5] In the forward direction, physical priors (such as upsampled point cloud initializations, mesh-coupled representations, and rigid-body solvers) improve geometric reconstruction quality.[4, 5] In the inverse direction, observed material dynamics are used to recover latent material parameters.[4, 5, 35]

Inverse Parameter Identification and Fracture Dynamics

This bidirectional coupling has been demonstrated on complex material behaviors, such as the fracturing of food materials.[4, 5] Under this framework, observed fracture patterns are analyzed using surrogate modeling and reinforcement learning to estimate latent material properties, such as tensile strength, fracture toughness, and plasticity limits.[4, 5] These estimated parameters are then fed back into an MPM simulator to generate physically plausible fracture animations under novel forces.[4, 5]

To support complex physical environments, AS-DiffMPM implements a differentiable MPM solver that handles contact with arbitrarily shaped colliders (beyond simple planar surfaces).[36] This allows the system to reconstruct shape and estimate material parameters (e.g., fluid viscosity) simultaneously as the material deforms and interacts with the environment.[36] Similarly, Resonance4D couples 3DGS with MPM to perform full-parameter material estimation.[37] It uses a zero-shot text-prompted segmentation model with simulation-guided initialization to segment reconstructed scenes into semantic object-part regions, supporting joint optimization of spatially varying material properties.[37]

Quantitative Performance and Convergence Comparison

By enforcing physical laws, physics-informed frameworks improve both dynamic view synthesis and geometric reconstruction accuracy compared to purely data-driven baselines.[9, 22]

Framework	Baseline Comparison	Primary Metric Improvement	Secondary Metric Improvement	Core Operational Advantage

PersistGS [22, 27]	Constant Velocity Extrapolation [22, 27]	+2.46 dB rendering PSNR under occlusion [22, 27]	40% reduction in trajectory estimation error [22, 27]	Maintains object tracking and permanence through complete visual occlusions [22, 27]
ReconPhys [9]	4DGS [9]	Chamfer Distance (CD) reduced from 0.593 to 0.001 [9]	Instant feedforward estimation (under 1 s) [9]	Eliminates the need for iterative per-scene optimization for asset creation [9]
i-PhysGaussian [18, 19]	Explicit MPM Baselines [18, 19]	Stable execution at up to 20× larger time steps [18, 19]	Preserved structural coherence under high stiffness [18, 19]	Prevents geometric collapse and numerical instability in stiff solids [18, 19]
PhysGaussian [12]	Fixed / Rigid Covariance Baselines [12]	Improved rendering quality (e.g., Ours: 26.96 dB PSNR) [12]	Accurate mechanical strain modeling [12]	Eliminates visual tearing and artifacts during large deformations [12]

GMR (Gaussian Mesh Renderer) [3]	Traditional Mesh Rasterizers [3]	5.26× faster average optimization speed [38]	Smoother analytical gradients with small batch sizes [3]	Highly memory-efficient surface reconstruction [3]
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Theoretical and Operational Bottlenecks

Despite the advancements in coupling visual and physical representations within the 3DGS framework, several theoretical and operational challenges remain open research directions:

- **Geometric Distortion and Elongation under Large Strain:** When Gaussian primitives are updated using the deformation gradient ($\Sigma_{\{t\}} = F_{\{t\}} \Sigma_{\{0\}} F_{\{t\}}^{\top}$), extreme stretching or compression can cause the ellipsoids to become excessively thin or elongated. [11, 14, 22] This violates the local-density assumptions of the splatting pipeline, producing noticeable "spiky" artifacts and visual noise. [11, 14, 22] To prevent this, models must implement anisotropic regularization terms or adaptive splitting and pruning schemes, which add computational complexity and can disrupt conservation laws. [11, 14, 17]
- **Gradient Vanishing in Specular and Reflective Scenes:** Standard visual-physical pipelines assume view-independent, Lambertian surface reflections.[7, 39] When capturing highly reflective or specular objects, the high-frequency nature of specular reflections makes them difficult to fit with low-order spherical harmonics.[39] Because lookup tables (such as environment maps) lack continuous, long-range gradients relative to surface normals, backpropagating rendering losses fails to provide useful gradients to update the physical geometry.[39]
- **Temporal Gradient Dissipation and Memory Scaling:** Backpropagating gradients through long physical trajectories (e.g., in fluid flow or multi-body contact simulations) requires storing intermediate states for every time step of the forward simulation.[40, 41] This direct backpropagation incurs high memory costs and can suffer from vanishing or exploding gradients over long sequences.[40, 41] Adjoint state methods, such as those utilizing bidirectional flow maps to backpropagate gradients along advection trajectories, are being explored to enable stable, long-range differentiation without storing intermediate solver variables.[41]

Summary

The emergence of differentiable, physics-informed 3D Gaussian Splatting represents a key development in unifying computer graphics with physical simulation. Endowing visual rendering primitives with physical attributes (such as mass, velocity, strain, and stress) allows simulations to run directly on the graphics primitives, adhering to the principle of "What You See Is What You Simulate." End-to-end differentiability enables the backpropagation of 2D rendering losses through physical solvers to estimate material properties, track objects through visual occlusions, and optimize functional shapes for target physical behaviors. While challenges in handling extreme deformations, specular reflections, and long-range gradient propagation remain, this

bidirectional visual-physical coupling provides a robust foundation for interactive digital twin creation, robotic manipulation, and functional physical design.

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